

Plots with Feature Values

15/11/2023

Load libraries

If the libraries are not installed yet, you need to install them using, for example, the command: `install.packages("ggplot2")`.

```
library(stringr)
library(ggplot2)
library(ggrepel)
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##   filter, lag

## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union

library(ggExtra)
library(ggpubr)
```

Load Data

Load csv file with quantitative feature values for samples per row. Comparative sample: A comparative sample of feature values for languages across the world, as well as other symbolic and non-symbolic sign sequences, is loaded here too. Note that the feature values were calculated as part of another Github project on “Natural Language Fingerprints” (NaLaFi), and published in Bentz (2023).

```
upperPal <- read.csv("~/Github/UpperPalCombinatorics/results/features.csv")
# select columns
upperPal <- select(upperPal, c('period', 'num.char', 'huni.chars', 'hrate.chars', 'ttr.chars', 'rm.chars'))
# change column names
colnames(upperPal) <- c('subcorpus', 'num.char', 'huni.chars', 'hrate.chars', 'ttr.chars', 'rm.chars')

# read feature values for languages and other sign systems
nalafi <- read.csv("~/Github/NaLaFi/results/features.csv")
# select columns
nalafi <- select(nalafi, c('subcorpus', 'num.char', 'huni.chars', 'hrate.chars', 'ttr.chars', 'rm.chars'))
# select only data with 100 chars
nalafi <- nalafi[nalafi$num.char == 100, ]

# combine the data frames
```

```
estimations.df <- rbind(upperPal, nalafi)
nrow(estimations.df)
```

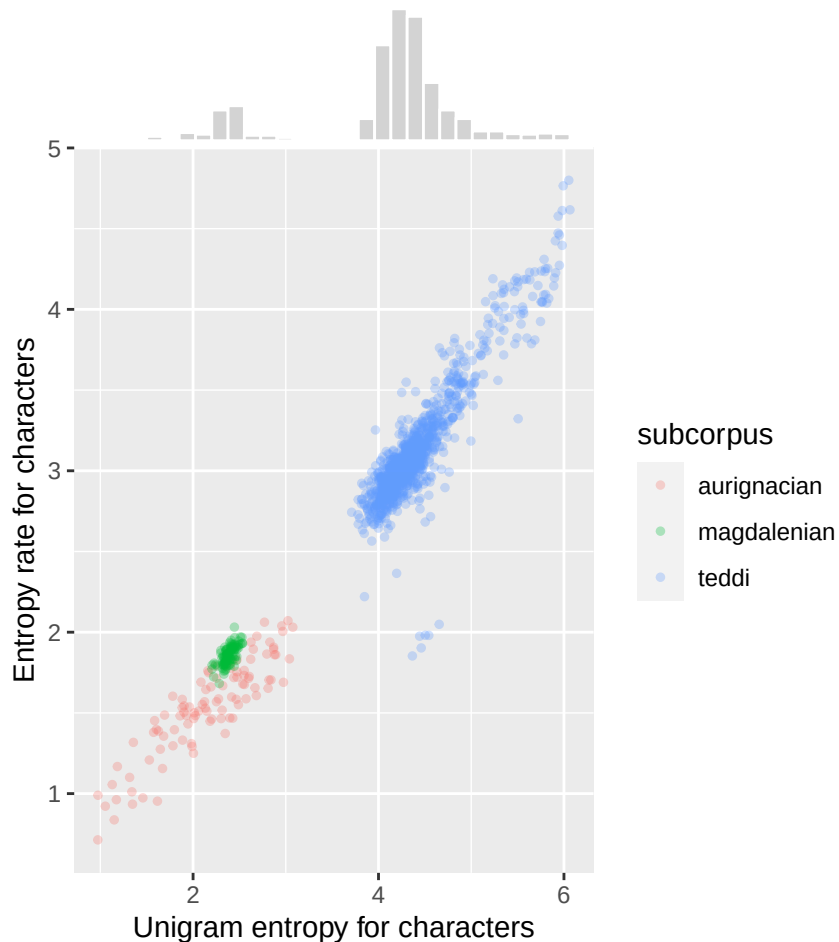
```
## [1] 3423
```

```
# select only certain subcorpora for plotting
```

```
estimations.df.selected <- estimations.df[estimations.df$subcorpus == "aurignacian"|
  estimations.df$subcorpus == "magdalenian"|
  estimations.df$subcorpus == "teddi", ]
```

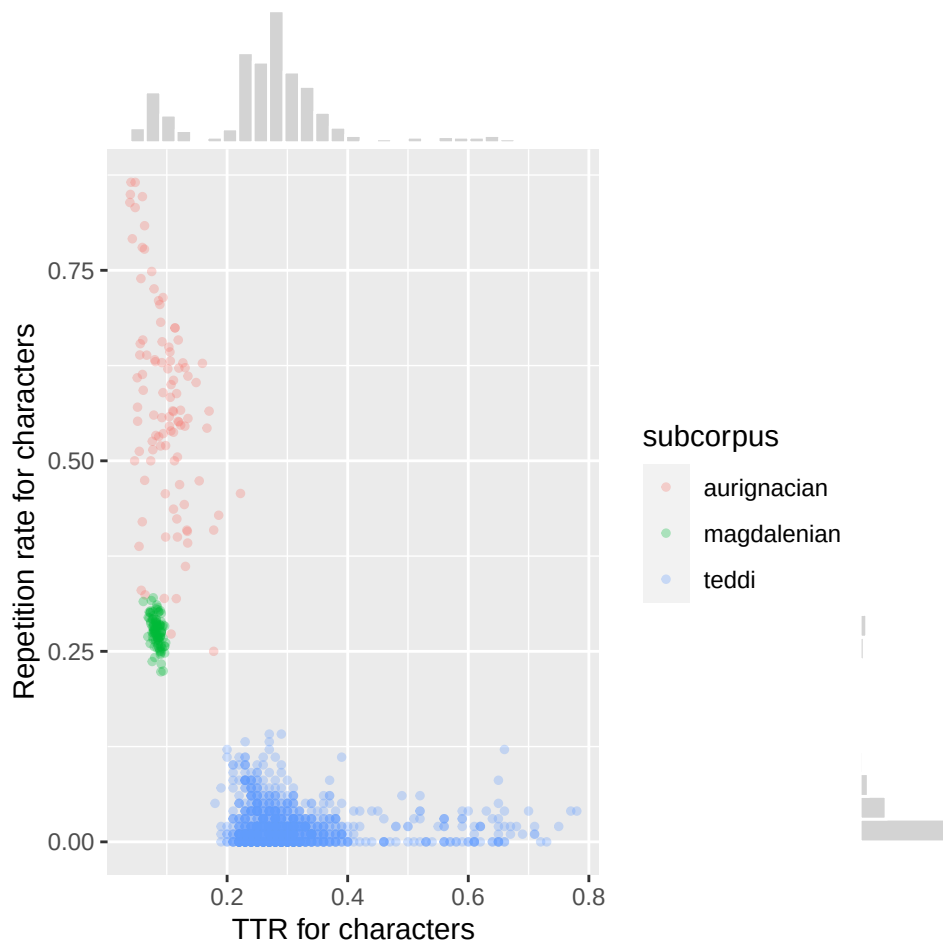
Entropy rate vs. unigram entropy

```
huni.hrate.chars.plot <- ggplot(estimations.df.selected, aes(x = huni.chars, y = hrate.chars,
  colour = subcorpus, fill = subcorpus)) +
  geom_point(alpha = 0.3, size = 1) +
  labs(x = "Unigram entropy for characters", y = "Entropy rate for characters") #+
  #theme(legend.position = c(.8,.2))
  #facet_wrap(~subcorpus)
huni.hrate.chars.plot <- ggMarginal(huni.hrate.chars.plot, type = "histogram", colour = "white",
  fill = "light grey")
huni.hrate.chars.plot
```



TTR vs. repetition rate

```
ttr.rm.chars.plot <- ggplot(estimations.df.selected, aes(x = ttr.chars, y = rm.chars,
                                                         colour = subcorpus)) +
  geom_point(alpha = 0.3, size = 1) +
  labs(x = "TTR for characters", y = "Repetition rate for characters") #+
  #theme(legend.position = c(.8,.2))
  #facet_wrap(~subcorpus)
ttr.rm.chars.plot <- ggMarginal(ttr.rm.chars.plot, type = "histogram", colour = "white",
                                fill = "light grey")
ttr.rm.chars.plot
```



Save complete figure to file

```
ggsave("~/Github/UpperPalCombinatorics/figures/huni_hrate_chars.pdf", huni.hrate.chars.plot, dpi = 300,
```